Solve with us 6.1 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

- A map $T: V \rightarrow W$T : V → W between two vector spaces VV and WW is said to be a linear transformation if for any two vectors v_1v1 and v_2v2 in the vector space VV and for any scalar cc in RR the following conditions hold:

1. $T(v_1 + v_2) = T(v_1) + T(v_2)$T(v1 + v2) = T(v1) + T(v2)
2. $T(cv_1) = cT(v_1)$T(cv1) = cT(v1)

1 point

Let TT be a function from R^2R2 to R^2R2 defined as follows:

$$T: R^2 \rightarrow R^2 \text{T : R2 → R2}$$
$$T(x, y) = (2x, y) \text{T(x, y) = (2x, y)}$$

Choose the set of correct option(s).
[Hint: Assume $v_1 = (x_1, y_1)$v1 = (x1, y1) and $v_2 = (x_2, y_2)$v2 = (x2, y2), then check the properties of linear transformation for the given function TT.]

- ☐ $T(v_1 + v_2) = T(v_1) + T(v_2)$T(v1 + v2) = T(v1) + T(v2), for all $v_1, v_2 \in R^2$ v1, v2 ∈ R2.
- ☐ $T(cv_1) = cT(v_1)$T(cv1) = cT(v1) for all $v_1 \in R^2$ v1 ∈ R2, and $c \in R$c ∈ R.
- ☐ TT is a linear transformation.
- ☐ TT is not a linear transformation.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(v_1 + v_2) = T(v_1) + T(v_2)$T(v1 + v2) = T(v1) + T(v2), for all $v_1, v_2 \in R^2$ v1, v2 ∈ R2.
$T(cv_1) = cT(v_1)$T(cv1) = cT(v1) for all $v_1 \in R^2$ v1 ∈ R2, and $c \in R$c ∈ R.
TT is a linear transformation.

1 point

Let TT be a function from R^2R2 to R^2R2 defined as follows:

$$T: R^2 \rightarrow R^2 \text{T : R2 → R2}$$
$$T(x, y) = (x - 1, y + 1) \text{T(x, y) = (x − 1, y + 1)}$$

Choose the set of correct option(s).
[Hint: Assume $v_1 = (x_1, y_1)$v1 = (x1, y1) and $v_2 = (x_2, y_2)$v2 = (x2, y2), then check the properties of linear transformation for the given function TT.]

- ○ $T(v_1 + v_2) = T(v_1) + T(v_2)$T(v1 + v2) = T(v1) + T(v2), for all $v_1, v_2 \in R^2$ v1, v2 ∈ R2.
- ○ $T(cv_1) = cT(v_1)$T(cv1) = cT(v1) for all $v_1 \in R^2$ v1 ∈ R2, and $c \in R$c ∈ R.
- ○ TT is a linear transformation.
- ○ TT is not a linear transformation.

No, the answer is incorrect.
Score: 0
Accepted Answers:
TT is not a linear transformation.

Key Points: 2Key Points: 2

- A linear transformation $T: V \rightarrow W$T : V → W is said to be an isomorphism if it is bijective, i.e. the followings hold:

1. $T(v_1) = T(v_2)$T(v1) = T(v2) implies $v_1 = v_2$ v1 = v2 (Injectivity).
2. For any $w \in W$ w ∈ W there exists $v \in V$ v ∈ V such that $T(v) = w$T(v) = w (Surjectivity).

1 point
If $T: R^2 \rightarrow R^2$T : R2 → R2 is a linear transformation, such that $T(x, y) = (x, 0)$T(x, y) = (x, 0), then which of the following options is true?
[[Hints:

Hint 1:Hint 1: Let $v_1 = (x_1, y_1)$v1 = (x1, y1), $v_2 = (x_2, y_2)$v2 = (x2, y2), and assume that $T(v_1) = T(v_2)$T(v1) = T(v2). Then check whether $v_1 = v_2$ v1 = v2 or not.
Hint 2:Hint 2: Does there exist any $v \in R^2$ v ∈ R2 for which $T(v) = (0, 1)$?]T(v) = (0, 1)?]

○ TT is both one to one and onto.
○ TT is one to one but not onto.
○ TT is onto but not one to one.
○ TT is neither one to one nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
TT is neither one to one nor onto.

1 point
If $T: R^2 \rightarrow R^3$T : R2 → R3 is a linear transformation defined by $T(x, y) = (2x + 3y, 5x - y, x + 6y)$T(x, y) = (2x + 3y, 5x − y, x + 6y). Which of the following options is true?
[Hint: Does there exist any $v \in R^2$ v ∈ R2 for which $T(v) = (1, 0, 0)$T(v) = (1, 0, 0)?]

○ TT is both one to one and onto.
○ TT is one to one but not onto.
○ TT is onto but not one to one.
○ TT is neither one to one nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
TT is one to one but not onto.

Key Points: 3Key Points: 3

- Standard Ordered basis of R^n Rn:

$$\{e_i \mid i = 1, 2, \ldots, n\}, \{\text{ei} \mid \text{i} = 1, 2, \ldots, \text{n}\},$$

where $e_1 = (1, 0, 0, \ldots, 0)$e1 = (1, 0, 0, … , 0), $e_2 = (0, 1, 0, \ldots, 0)$e2 = (0, 1, 0, … , 0), … …, $e_n = (0, 0, \ldots, 0, 1)$en = (0, 0, … , 0, 1). In general, $e_i = (0, 0, \ldots, 0, 1, 0, \ldots, 0)$ei = (0, 0, … , 0, 1, 0, … , 0), where 11 is at the ii-th position and 00 elsewhere.

- If $T(e_i)$T(ei) is known for all ii, then the description of TT can be known entirely. Suppose $T(e_i) = v_i$ T(ei) = vi, for $i = 1, 2, \ldots, n$i = 1, 2, … , n, then we have

$T(x_1, x_2, \ldots, x_n) = T(x_1 e_1 + x_2 e_2 + \ldots + x_n e_n)$T(x1, x2, … , xn) = T(x1e1 + x2e2 + … + xn en)

$= T(x_1 e_1) + T(x_2 e_2) + \ldots + T(x_n e_n) = x_1 T(e_1) + x_2 T(e_2) + \ldots + x_n T(e_n) = x_1 v_1 + x_2 v_2$ = T(x1e1) + T(x2e2) + … + T(xnen)= x1T(e1) + x2T(e2) + … + xnT(en)= x1v1 + x2v2 + … + xnvn

1 point

Let $T: R^2 \to R^2$ T : R2 → R2 be a linear transformation, such that $T(1, 0) = (3, 2)$T(1, 0) = (3, 2) and $T(0, 1) = (1, 5)$T(0, 1) = (1, 5). Which of the following options is true?

[Hint: $\{(1, 0), (0, 1)\}${(1, 0), (0, 1)} is the standard ordered basis of R^2 R2.]

○ $T(x, y) = (3x + y, 2x + 5y)$T(x, y) = (3x + y, 2x + 5y)

○ $T(x, y) = (3x + 2y, x + 5y)$T(x, y) = (3x + 2y, x + 5y)

○ $T(x, y) = (5x, 6y)$T(x, y) = (5x, 6y)

○ $T(x, y) = (4x, 7y)$T(x, y) = (4x, 7y)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y) = (3x + y, 2x + 5y)$T(x, y) = (3x + y, 2x + 5y)

1 point

Let $T: R^2 \to R^2$ T : R2 → R2 be a linear transformation, such that $T(2, 1) = (3, 2)$T(2, 1) = (3, 2) and $T(1, 3) = (0, 0)$T(1, 3) = (0, 0). Which of the following options is true?

[Hint: Express (1, 0)(1, 0) and (0, 1)(0, 1) in terms of (2, 1)(2, 1) and (1, 3)(1, 3).]

○ $T(x, y) = \frac{1}{5}(3x, 2y)$T(x, y) = 51(3x, 2y)

○ $T(x, y) = (x + 1, y + 1)$T(x, y) = (x + 1, y + 1)

○ $T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$T(x, y) = 51(9x − 3y, 6x − 2y)

○ TT cannot be determined from the given information.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:
$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$T(x, y) = 51(9x − 3y, 6x − 2y)

Key Points: 4Key Points: 4

- Let $T: V \rightarrow W$T : V → W be a linear transformation. Let $\beta = \{v_1, v_2, ..., v_n\}$β = {v1, v2, ..., vn} be an ordered basis of VV and $\gamma = \{w_1, w_2, ..., w_m\}$γ = {w1, w2, ..., wm} be an ordered basis of WW. So each $f(v_i)$f(vi) can be uniquely written as linear combination of w_j wj's, where $i = 1, 2, ..., n$ i = 1, 2, ..., n and $j = 1, 2, ..., m$j = 1, 2, ..., m.

$T(v_1) = a_{11} w_1 + a_{21} w_2 + ... + a_{m1} w_m$ $T(v_2) = a_{12} w_1 + a_{22} w_2 + ... + a_{m2} w_m$ ⋮ $T(v_n) = a_{1n} w_1$ T(v1) = a11w1 + a21w2 + ... + am1wmT(v2) = a12w1 + a22w2 + ... + am2wm ⋮ T(vn) = a1nw1 + a2nw2 + ... + amnwm

The matrix corresponding to the linear transformation TT with respect to the ordered bases ββ and γγ is given by,

$$\begin{bmatrix} a_{11} & a_{12} & ... & a_{1n} \\ a_{21} & a_{22} & ... & a_{2n} \\ \vdots & \vdots & & \\ a_{m1} & a_{m2} & ... & a_{mn} \end{bmatrix}$$ a11 a12...a1na21 a22...a2n ⋮ ⋮ am1 am2...amn

The above matrix not only depends on the linear map TT, but also on the choice of the ordered bases ββ for VV and γγ for WW.

- It is important to note that, for a fixed choice of basis of VV and WW, there is a one to one correspondence between the set of all linear transformations from VV to WW, and the set of all $m \times n$ m × n matrices, where $n = \dim(V)$n = dim(V) and $m = \dim(W)$m = dim(W).

1 point

Suppose $T: R^2 \rightarrow R^2$ T : R2 → R2 is a linear transformation, such that $T(x, y) = (x, 0)$T(x, y) = (x, 0). Which of the following matrices corresponds to TT with respect to the standard ordered basis of R^2 R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for both the domain and co-domain?

[Hint: $T(1, 0) = (1, 0)$T(1, 0) = (1, 0) and $T(0, 1) = (0, 0)$T(0, 1) = (0, 0)]

○ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$[1001]

○ $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$[1100]

○

$\begin{bmatrix}1&0\\0&0\end{bmatrix}$[1000]

○
$\begin{bmatrix}1&0\\-1&1\end{bmatrix}$[1−101]

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix}1&0\\0&0\end{bmatrix}$[1000]

1 point
Suppose $T: R^2 \rightarrow R^2$ T : R2 → R2 is a linear transformation, such that $T(x, y) = (x + y, x - y)$ T (x, y) = (x + y, x − y). Which of the following matrices corresponds to T T with respect to the standard ordered basis of R^2 R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for the domain and $\{(1, 1), (1, -1)\}$ {(1, 1), (1, −1)} for the co-domain?
[Hint: $T(1, 0) = (1, 1)$ T (1, 0) = (1, 1) and $T(0, 1) = (1, -1)$ T (0, 1) = (1, −1)]
○
$\begin{bmatrix}1&1\\-1&1\end{bmatrix}$[1−111]

○
$\begin{bmatrix}1&0\\0&-1\end{bmatrix}$[100−1]

○
$\begin{bmatrix}1&0\\0&1\end{bmatrix}$[1001]

○
$\begin{bmatrix}1&-1\\1&1\end{bmatrix}$[11−11]

No, the answer is incorrect.
Score: 0
Feedback:
• T(1, 0) = (1, 1) = 1(1, 1) + 0(1, −1), and T(0, 1) = (1, −1) = 0(1, 1) + 1(1, −1).
Accepted Answers:
$\begin{bmatrix}1&0\\0&1\end{bmatrix}$[1001]

1 point
Let $W = \{(x, y, z) \mid x = 2y + z\}$ W = {(x, y, z) ∣ x = 2y + z} be a subspace of R^3 R3. Let $\beta = \{(2, 1, 0), (1, 0, 1)\}$ β = {(2, 1, 0), (1, 0, 1)} be a basis of W W. Let $T: W \rightarrow R^2$ T : W → R2 be a linear transformation such that $T(2, 1, 0) = (1, 0)$ T (2, 1, 0) = (1, 0) and $T(1, 0, 1) = (0, 1)$ T (1, 0, 1) = (0, 1). What will be the matrix representation of T T with respect to the basis β β for W W and $\gamma = \{(1, 1), (1, -1)\}$ γ = {(1, 1), (1, −1)} for R^2 R2?
○
$\frac{1}{2}\begin{bmatrix}1&1\\1&-1\end{bmatrix}$21 [111−1]

○
$\frac{1}{2}\begin{bmatrix}1&0\\0&1\end{bmatrix}$21 [1001]

○
$\begin{bmatrix}1&1\\1&-1\end{bmatrix}$[111−1]

○
$\begin{bmatrix}1&1\\-1&1\end{bmatrix}$[1−111]

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$21 [111−1]

1 point
Let $\beta = \{(1,0),(0,1)\}$β = {(1, 0), (0, 1)} and $\gamma = \{(1,1),(1,-1)\}$γ = {(1, 1), (1, −1)} be two bases of R^2 R2. If $\begin{bmatrix}a & b\\ c & d\end{bmatrix}$[acbd] is the matrix representation of a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 with respect to ββ for the both the domain and co-domain. What will be the matrix representation of T T with respect to γγ for the domain and ββ for the co-domain?
[Hint: First find $T(x,y)$T(x, y) and then find $T(1,1)$T(1, 1) and $T(1,-1)$T(1, −1).]

- $\begin{bmatrix}a+b & c+d\\ a-b & c-d\end{bmatrix}$[a + ba − bc + dc − d]
- $\begin{bmatrix}a-b\\ c-d\end{bmatrix}$[ac−b−d]
- $\begin{bmatrix}a+b & a-b\\ c+d & c-d\end{bmatrix}$[a + bc + da − bc − d]
- $\begin{bmatrix}a+b & -a-b\\ c+d & -c-d\end{bmatrix}$[a + bc + d−a − b−c − d]

No, the answer is incorrect.
Score: 0
Feedback:
• T(x, y) = (ax+by, cx+dy), so T(1, 1) = (a+b, c+d) and T(1, −1) = (a−b, c−d).
Accepted Answers:
$\begin{bmatrix}a+b & a-b\\ c+d & c-d\end{bmatrix}$[a + bc + da − bc − d]